

Commerce XI Maths App

Activities 3.1–3.2

Detailed Answers with Explanations and Steps

Mathematics & Statistics, Part 1, Unit 3 - Complex Numbers | Total questions: 21

Question 1 - MCQ · PG-43 · ACT-3.1 · Step 1

If $i = \sqrt{-1}$, find the value of i^6

Options

- A. -1
- B. 1
- C. i
- D. $-i$

Correct answer: -1

Explanation and steps

This is the stated step in solving Activity 3.1. Use $i^2 = -1$, equate real and imaginary parts when required, and simplify to obtain -1.

Question 2 - MCQ · PG-43 · ACT-3.1 · Step 2

If $i = \sqrt{-1}$, find the value of i^3

Options

- A. $-i$
- B. i
- C. 1
- D. -1

Correct answer: $-i$

Explanation and steps

This is the stated step in solving Activity 3.1. Use $i^2 = -1$, equate real and imaginary parts when required, and simplify to obtain $-i$.

Question 3 - MCQ · PG-43 · ACT-3.1 · Step 3

If $i^6 = -1$, find the coefficient of y after simplifying $-15i^6y$

Options

- A. 15
- B. -15
- C. $15i$
- D. $-15i$

Correct answer: 15

Explanation and steps

This is the stated step in solving Activity 3.1. Use $i^2 = -1$, equate real and imaginary parts when required, and simplify to obtain 15.

Question 4 - MCQ · PG-43 · ACT-3.1 · Step 4

If $i^4 = -i$, simplify $i^4(y + 4)$

Options

- A. $-(y + 4)i$
- B. $(y + 4)i$
- C. $-i(y - 4)$
- D. $y + 4$

Correct answer: $-(y + 4)i$

Explanation and steps

This is the stated step in solving Activity 3.1. Use $i^2 = -1$, equate real and imaginary parts when required, and simplify to obtain $-(y + 4)i$.

Question 5 - MCQ · PG-43 · ACT-3.1 · Step 5

Simplify the equation

$$x + 2i = 7x - 15i^6y + i^3(y + 4),$$

using $i^6 = -1$ and $i^3 = -i$

Options

- A. $x - 15y + 2i = 7x - (y + 4)i$
- B. $x + 15y + 2i = 7x + (y + 4)i$
- C. $x - 15y + 2i = 7x + (y + 4)i$
- D. $x + 15y + 2i = 7x - (y + 4)i$

Correct answer: $x - 15y + 2i = 7x - (y + 4)i$

Explanation and steps

This is the stated step in solving Activity 3.1. Use $i^2 = -1$, equate real and imaginary parts when required, and simplify to obtain $x - 15y + 2i = 7x - (y + 4)i$.

Question 6 - MCQ · PG-43 · ACT-3.1 · Step 6

$$\text{In } x - 15y + 2i = 7x - (y + 4)i,$$

the coefficient of i , on the left side is

Options

- A. 2
- B. -2
- C. 0
- D. i

Correct answer: 2

Explanation and steps

This is the stated step in solving Activity 3.1. Use $i^2 = -1$, equate real and imaginary parts when required, and simplify to obtain 2.

Question 7 - MCQ · PG-43 · ACT-3.1 · Step 7

Equating the real parts of $x - 15y + 2i = 7x - (y + 4)i$ gives

Options

- A. $-6x - 15y = 0$
- B. $-6x + 15y = 0$
- C. $6x - 15y = 2$
- D. $-6x - 15y = 2$

Correct answer: $-6x - 15y = 0$

Explanation and steps

This is the stated step in solving Activity 3.1. Use $i^2 = -1$, equate real and imaginary parts when required, and simplify to obtain $-6x - 15y = 0$.

Question 8 - MCQ · PG-43 · ACT-3.1 · Step 8

Equating the imaginary parts of $x - 15y + 2i = 7x - (y + 4)i$ gives

Options

- A. $2 = -(y + 4)$
- B. $-2 = -(y + 4)$
- C. $2 = y + 4$
- D. $0 = -(y + 4)$

Correct answer: $2 = -(y + 4)$

Explanation and steps

This is the stated step in solving Activity 3.1. Use $i^2 = -1$, equate real and imaginary parts when required, and simplify to obtain $2 = -(y + 4)$.

Question 9 - MCQ · PG-43 · ACT-3.1 · Step 9

Simplify the equation $x - 15y = 7x$

Options

- A. $-6x - 15y = 0$
- B. $-6x + 15y = 0$
- C. $6x - 15y = 0$
- D. $-6x - 15y = 2$

Correct answer: $-6x - 15y = 0$

Explanation and steps

This is the stated step in solving Activity 3.1. Use $i^2 = -1$, equate real and imaginary parts when required, and simplify to obtain $-6x - 15y = 0$.

Question 10 - MCQ · PG-43 · ACT-3.1 · Step 10

Solve $2 = -(y + 4)$ for y

Options

- A. -6
- B. 6
- C. -2
- D. 2

Correct answer: -6

Explanation and steps

This is the stated step in solving Activity 3.1. Use $i^2 = -1$, equate real and imaginary parts when required, and simplify to obtain -6.

Question 11 - MCQ · PG-43 · ACT-3.1 · Step 11

If $y = -6$ and $-6x - 15y = 0$, find x

Options

- A. 15
- B. -15
- C. 9
- D. 6

Correct answer: 15

Explanation and steps

This is the stated step in solving Activity 3.1. Use $i^2 = -1$, equate real and imaginary parts when required, and simplify to obtain 15.

Question 12 - MCQ · PG-43 · ACT-3.1 · Step 12

Solve the equations $-6x - 15y = 0$ and $2 = -(y + 4)$. The ordered pair (x, y) is

Options

- A. (15, -6)
- B. (-15, 6)
- C. (9, -6)
- D. (15, 6)

Correct answer: (15, -6)

Explanation and steps

This is the stated step in solving Activity 3.1. Use $i^2 = -1$, equate real and imaginary parts when required, and simplify to obtain (15, -6).

Question 13 - MCQ · PG-43 · ACT-3.1 · Step 13

If $x = 15$ and $y = -6$, find $x + y$

Options

- A. 9
- B. -9
- C. 21
- D. -21

Correct answer: 9

Explanation and steps

This is the stated step in solving Activity 3.1. Use $i^2 = -1$, equate real and imaginary parts when required, and simplify to obtain 9.

Question 14 - MCQ · PG-43 · ACT-3.2 · Step 1

If $w^3 = 1$, which factor multiplied by $c + aw + bw^2$ gives $a + bw + cw^2$?

Options

- A. w^2
- B. w
- C. w^3
- D. $(1)/(w^2)$

Correct answer: w^2

Explanation and steps

Use $w^3 = 1$ together with $1/w = w^2$ and $1/w^2 = w$. Substitution in this step gives w^2 .

Question 15 - MCQ · PG-43 · ACT-3.2 · Step 2

Complete the equivalent fractions: $(a)/(w^2) = (a \text{ ___})/(w^3)$

Options

- A. w
- B. w^2
- C. 1
- D. $-w$

Correct answer: w

Explanation and steps

Use $w^3 = 1$ together with $1/w = w^2$ and $1/w^2 = w$. Substitution in this step gives w .

Question 16 - MCQ · PG-43 · ACT-3.2 · Step 3

Complete the equivalent fractions: $(b)/(w) = (b \text{ ___ })/(w^3)$

Options

- A. w^2
- B. w
- C. 1
- D. $-w^2$

Correct answer: w^2

Explanation and steps

Use $w^3 = 1$ together with $1/w = w^2$ and $1/w^2 = w$. Substitution in this step gives w^2 .

Question 17 - MCQ · PG-43 · ACT-3.2 · Step 4

If $w^3 = 1$, simplify $(a)/(w^2)$

Options

- A. aw
- B. aw^2
- C. a
- D. $(a)/(w)$

Correct answer: aw

Explanation and steps

Use $w^3 = 1$ together with $1/w = w^2$ and $1/w^2 = w$. Substitution in this step gives aw .

Question 18 - MCQ · PG-43 · ACT-3.2 · Step 5

If $w^3 = 1$, simplify $(b)/(w)$

Options

- A. bw^2
- B. bw
- C. b
- D. $(b)/(w^2)$

Correct answer: bw^2

Explanation and steps

Use $w^3 = 1$ together with $1/w = w^2$ and $1/w^2 = w$. Substitution in this step gives bw^2 .

Question 19 - MCQ · PG-43 · ACT-3.2 · Step 6

If $w^3 = 1$, simplify $(a)/(w^2) + (b)/(w) + c$

Options

- A. $aw + bw^2 + c$
- B. $aw^2 + bw + c$
- C. $a + b + c$
- D. $a + bw + cw^2$

Correct answer: $aw + bw^2 + c$

Explanation and steps

Use $w^3 = 1$ together with $1/w = w^2$ and $1/w^2 = w$. Substitution in this step gives $aw + bw^2 + c$.

Question 20 - MCQ · PG-43 · ACT-3.2 · Step 7

In the fraction $(aw + bw^2 + c)/(c + aw + bw^2)$, the numerator and denominator are

Options

- A. equal
- B. additive inverses
- C. conjugates
- D. reciprocals

Correct answer: equal

Explanation and steps

Use $w^3 = 1$ together with $1/w = w^2$ and $1/w^2 = w$. Substitution in this step gives equal.

Question 21 - MCQ · PG-43 · ACT-3.2 · Step 8

If $w^3 = 1$, find $(a + bw + cw^2)/(c + aw + bw^2)$ in terms of w^2

Options

- A. w^2
- B. w
- C. 1
- D. $-w^2$

Correct answer: w^2

Explanation and steps

Use $w^3 = 1$ together with $1/w = w^2$ and $1/w^2 = w$. Substitution in this step gives w^2 .